

RecoMart Recommendation Pipeline - Problem Formulation Report

Business Problem

RecoMart needs a scalable recommendation pipeline that improves conversion and cross-selling by learning from fresh user interactions, purchases, product metadata, and external product signals.

Objectives

Create clean EDA-ready datasets, engineered features for collaborative and content-based recommenders, a deployable recommendation model, and an inference interface for top-K recommendations.

Data Sources

CSV clickstream/interactions: user_id, item_id, event_type, rating, quantity, price, event_ts, channel. REST API product catalog: item_id, category, brand, price, sentiment_score, popularity_score.

Expected Outputs

Raw immutable data lake partitions, quality reports, prepared datasets, transformed feature tables, a versioned feature store, trained SVD model, evaluation metrics, and orchestration logs.

Evaluation Metrics

Precision@K measures the share of recommended items that are relevant. Recall@K measures coverage of relevant items. NDCG@K rewards placing relevant items near the top of the ranked list.